

Course Source Register

Primary literature and official documentation, with the teaching scope made explicit

Prepared for Yasin Kaymaz, PhD | Version 1.0 | 14 September 2026

This is a teaching package, not a clinical, legal or institutional accreditation document. Scientific sources are identified by [Sxx] in the Course Source Register.

How to use sources

Source IDs are printed on slides, in speaker notes and in the handbooks. URLs were checked during preparation on 14 September 2026. Database and software pages can change; preserve the relevant version or access date when reusing a fact. The source register is not a claim that every source is the latest clinical standard. Old foundational papers are used for their stated foundational contribution. Dynamic database summaries are snapshots, not immutable facts.

S01 | **Alberts et al. Molecular Biology of the Cell, 4th ed. Foundational cell biology chapters.**

Foundational textbook | DNA, RNA, proteins, genes and biological measurement. Older edition used only for stable foundations.

<https://www.ncbi.nlm.nih.gov/books/NBK21054/>

S02 | **GENCODE Human Release 50 and release-specific statistics.**

Official database documentation | Annotation counts depend on release and counting rules; do not memorize a universal gene count.

https://www.genencodegenes.org/human/stats_50.html

S03 | **SAM/BAM, VCF and BED specifications, maintained by the HTS community.**

Official specifications | Coordinate conventions and file contents. SAM text POS and BAM internal storage must not be conflated.

<https://samtools.github.io/hts-specs/>

S04 | **UCSC Genome Browser: data format FAQ.**

Official documentation | BED is zero-based, end-exclusive; other formats require their own specifications.

<https://genome.ucsc.edu/FAQ/FAQformat.html>

S05 | **Python 3 tutorial: introduction and control flow.**

Official documentation | Lists, indexing, functions, tests and execution state.

<https://docs.python.org/3/tutorial/>

S06 | **NCBI: How BLAST Works and BLAST Statistics: The Expect Value.**

Official training | An E-value is an expected count under a sequence-score model, not the probability of common function.

https://www.nlm.nih.gov/ncbi/workshops/2023-08_BLAST_evol/e_value.html

S07 | American Statistical Association. Statement on statistical significance and p-values (2016).

Professional statistical guidance | Interpretation of p-values, effect sizes and uncertainty.

<https://www.amstat.org/asa/files/pdfs/P-ValueStatement.pdf>

S08 | Benjamini and Hochberg (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing.

Primary statistical method | BH step-up adjustment; original guarantee under independence, extensions under specified dependence.

<https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>

S09 | Love, Huber and Anders (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2.

Primary method | Count-based differential expression, normalization and variation between samples.

<https://doi.org/10.1186/s13059-014-0550-8>

S10 | DESeq2: Analyzing RNA-seq data with DESeq2, release vignette.

Official software documentation | Count input; sample metadata alignment; count-model interpretation. Real two-gene excerpt used only for alignment literacy.

<https://bioconductor.org/packages/release/bioc/vignettes/DESeq2/inst/doc/DESeq2.html>

S11 | scikit-learn: Common pitfalls and recommended practices.

Official software documentation | Data leakage, splitting before preprocessing, fitting transformations on training data only.

https://scikit-learn.org/stable/common_pitfalls.html

S12 | Squair et al. (2021). Confronting false discoveries in single-cell differential expression.

Primary benchmark | Biological replication and between-replicate variation in condition comparisons.

<https://doi.org/10.1038/s41467-021-25960-2>

S13 | AnnData documentation: annotated data matrices.

Official software documentation | Convention: observations are rows, variables are columns. Layers must be inspected rather than presumed to be counts.

<https://anndata.readthedocs.io/en/stable/>

S14 | Cui et al. (2024). scGPT: toward building a foundation model for single-cell multi-omics using generative AI.

Primary research | An example of pretrained biological representations and task-specific adaptation; not a universal performance claim.

<https://doi.org/10.1038/s41592-024-02201-0>

S15 | Kedzierska et al. (2025). Zero-shot evaluation reveals limitations of single-cell foundation models.

Primary benchmark | Specific zero-shot evaluations; results must not be generalized to every foundation model or fine-tuning regime.

<https://doi.org/10.1186/s13059-025-03574-x>

S16 | EMBL-EBI AlphaFold training: evaluating predicted structures using confidence scores.

Official training | pLDDT and PAE have different meanings; confidence is not a binding assay.

<https://www.ebi.ac.uk/training/online/courses/alphafold/inputs-and-outputs/evaluating-alphafolds-predicted-structures-using-confidence-scores/>

S17 | Jumper et al. (2021). Highly accurate protein structure prediction with AlphaFold.

Primary research | Protein structure prediction and its evaluation; no equivalence to functional validation.

<https://doi.org/10.1038/s41586-021-03819-2>

S18 | NCBI ClinVar: review status and representation of classifications.

Official database documentation | Review status, conflicts and context; review stars are not probabilities of pathogenicity.

https://www.ncbi.nlm.nih.gov/clinvar/docs/review_status/

S19 | Richards et al. (2015). ACMG/AMP standards and guidelines for interpretation of sequence variants.

Foundational consensus guideline | Mendelian germline classification categories; not a complete current clinical training protocol. Consult current gene/disease specifications in practice.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC4544753/>

S20 | Turkish Personal Data Protection Law No. 6698, official English translation with 2024 amendments.

Official legal source | Genetic data and health data; conditions of processing. Turkish original prevails.

<https://www.kvkk.gov.tr/Icerik/6649/Personal-Data-Protection-Law>

S21 | KVKK: international transfers, Article 9 amendments and safeguards.

Official legal source | Cross-border transfers require an applicable route, not merely removal of names.

<https://www.kvkk.gov.tr/Icerik/2053/Yurtdisina-Aktarim>

S22 | Regulation (EU) 2016/679 (GDPR), official EUR-Lex text.

Official legal source | Genetic data, special categories, pseudonymization and transfers; applicability depends on context.

<https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng>

S23 | ICMJE recommendations: use of AI by authors.

Official publication guidance | Disclosure and human responsibility; publication guidance is not university assessment regulation.

<https://www.icmje.org/recommendations/browse/artificial-intelligence/ai-use-by-authors.html>

S24 | RCSB PDB computed model AF_AFP69905F1 / AlphaFold DB P69905.

Official structure record | Real evidence card: hemoglobin alpha, 142 modeled residues, displayed global pLDDT 98.06 at retrieval. Not experimental data.

https://www.rcsb.org/structure/AF_AFP69905F1

S25 | Huber and Reyes. pasilla data package; Brooks et al. (2011) experiment.

Official data provenance | RNAi perturbation in Drosophila. Only a two-gene published count excerpt is bundled; not a full differential-expression dataset.

<https://bioconductor.org/packages/release/data/experiment/html/pasilla.html>

S26 | NCBI: How BLAST Works (2022 workshop).

Official training | Local alignment and heuristic search.

https://www.nlm.nih.gov/ncbi/workshops/2022-10_Basic-Web-BLAST/how-blast-works.html

S27 | EMBL-EBI: Single cell RNA-seq analysis using Python materials (2026).

Official training | Optional next-step learning; not a required software installation.

<https://www.ebi.ac.uk/training/materials/single-cell-rna-seq-analysis-using-python-materials/>

S28 | Smith and Waterman (1981). Identification of common molecular subsequences. Journal of Molecular Biology 147:195-197.

Primary algorithm | Exact local alignment under a scoring rule; teaching implementation uses a linear gap cost.

[https://doi.org/10.1016/0022-2836\(81\)90087-5](https://doi.org/10.1016/0022-2836(81)90087-5)

Additional method source

S28 | Smith and Waterman (1981). Identification of common molecular subsequences. Journal of Molecular Biology 147:195-197. The W04 local-score code is an original minimal implementation of the linear-gap teaching recurrence, not production alignment software.

[https://doi.org/10.1016/0022-2836\(81\)90087-5](https://doi.org/10.1016/0022-2836(81)90087-5)

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